Supplementary Material: The Torus Electron

James P. Galasyn and Claude Théodore

1 S1. Torus Knot Parameterization

A (p, q) torus knot on a torus with major radius R and minor radius r is parameterized by $\lambda \in [0, 2\pi)$:

$$x(\lambda) = (R + r\cos q\lambda)\cos p\lambda, \tag{1}$$
$$y(\lambda) = (R + r\cos q\lambda)\sin p\lambda, \tag{2}$$
$$z(\lambda) = r\sin q\lambda. \tag{3}$$

The toroidal angle advances as $\theta = p\lambda$ and the poloidal angle as $\phi = q\lambda$. For $(p, q) = (2, 1)$, the curve winds twice toroidally for each poloidal circuit, closing after λ traverses $[0, 2\pi)$.

Tangent vectors. Differentiating:

$$\frac{dx}{d\lambda} = -rq\sin(q\lambda)\cos(p\lambda) - (R + r\cos(q\lambda))\,p\sin(p\lambda), \tag{4}$$
$$\frac{dy}{d\lambda} = -rq\sin(q\lambda)\sin(p\lambda) + (R + r\cos(q\lambda))\,p\cos(p\lambda), \tag{5}$$
$$\frac{dz}{d\lambda} = rq\cos(q\lambda). \tag{6}$$

Arc length element. After expanding and simplifying (cross terms cancel in $dx^2 + dy^2$):

$$\left|\frac{d\mathbf{r}}{d\lambda}\right|^2 = r^2q^2 + (R + r\cos q\lambda)^2p^2. \tag{7}$$

In the thin-torus limit $r \ll R$:

$$\left|\frac{d\mathbf{r}}{d\lambda}\right| \approx \sqrt{p^2R^2 + q^2r^2}\left(1 + \frac{Rrp^2\cos(q\lambda)}{p^2R^2 + q^2r^2}\right). \tag{8}$$

The cosine term integrates to zero over a full period, giving the total path length:

$$L = \int_0^{2\pi}\left|\frac{d\mathbf{r}}{d\lambda}\right| d\lambda \approx 2\pi\sqrt{p^2R^2 + q^2r^2}. \tag{9}$$

For $(2, 1)$ with $r = \alpha R$: $L \approx 2\pi R\sqrt{4 + \alpha^2} \approx 4\pi R$.

2 S2. Self-Energy Derivation

2.1 Neumann inductance

The magnetic self-energy of a current loop on a torus is determined by the Neumann inductance formula [1]. For a thin torus ($r \ll R$), the inductance of the toroidal current path is:

$$\mathcal{L} = \mu_0 R\left[\ln\frac{8R}{r} - 2\right]. \tag{10}$$

This is the standard result for a circular loop of radius R with wire radius r (Jackson §5.6 [1]).

2.2 Current and self-energy

A charge e traversing a closed path of length L at speed c creates a time-averaged current:

$$I = \frac{e}{T} = \frac{ec}{L}. \tag{11}$$

The magnetic self-energy is:

$$U_{\text{mag}} = \frac{1}{2}\mathcal{L}I^2 = \frac{\mu_0 R}{2}\left[\ln\frac{8R}{r} - 2\right]\left(\frac{ec}{L}\right)^2. \tag{12}$$

2.3 Electric–magnetic equality at $v = c$

For a charge moving at speed v, the electromagnetic field in the lab frame has energy densities:

$$u_B = \frac{B^2}{2\mu_0}, \qquad u_E = \frac{\varepsilon_0 E^2}{2}. \tag{13}$$

In the ultrarelativistic limit $v \to c$, the field becomes purely transverse with $|\mathbf{E}| = c|\mathbf{B}|$, giving $u_E = u_B$. The total electromagnetic self-energy is therefore:

$$U_{\text{EM}} = U_{\text{mag}} + U_{\text{elec}} = 2U_{\text{mag}} = \mu_0 R\left[\ln\frac{8R}{r} - 2\right]\left(\frac{ec}{L}\right)^2. \tag{14}$$

2.4 Ratio to circulation energy

The circulation energy is $E_{\text{circ}} = 2\pi\hbar c/L$. The ratio:

$$\frac{U_{\text{EM}}}{E_{\text{circ}}} = \frac{\mu_0 R[\ln(8R/r) - 2](ec)^2/L}{2\pi\hbar c}. \tag{15}$$

Using $L \approx 2\pi pR$ and $\mu_0 e^2 c = 4\pi\hbar\alpha$:

$$\frac{U_{\text{EM}}}{E_{\text{circ}}} = \frac{4\pi\hbar\alpha \cdot R \cdot [\ln(8R/r) - 2]}{2\pi pR \cdot 2\pi\hbar} = \frac{\alpha[\ln(8R/r) - 2]}{p\pi}. \tag{16}$$

For $p = 2$ and $r/R = \alpha$: $U_{\text{EM}}/E_{\text{circ}} = \alpha[\ln(8/\alpha) - 2]/(2\pi) \approx 0.0058$, or about 0.6%. The self-energy is a perturbative correction to the dominant circulation energy.

3 S3. Angular Momentum Computation

3.1 Instantaneous angular momentum

At parameter λ, the photon has position $\mathbf{r}(\lambda)$ and momentum $\mathbf{p} = (E_{\text{total}}/c)\hat{v}(\lambda)$, where $\hat{v} = \mathbf{v}/c$ is the unit tangent. The z-component of angular momentum is:

$$L_z(\lambda) = [\mathbf{r} \times \mathbf{p}]_z = \frac{E_{\text{total}}}{c}\left[x\hat{v}_y - y\hat{v}_x\right]. \tag{17}$$

3.2 Time average

The photon moves at constant speed c, so it spends equal proper time per unit arc length. The time-averaged angular momentum is:

$$\langle L_z \rangle = \frac{\int_0^{2\pi} L_z(\lambda)\left|\frac{d\mathbf{r}}{d\lambda}\right| d\lambda}{\int_0^{2\pi}\left|\frac{d\mathbf{r}}{d\lambda}\right| d\lambda}. \tag{18}$$

This is evaluated numerically with $N = 2000$–4000 sample points.

3.3 Sweep over r/R

Holding R fixed and sweeping r/R from 0 to 1:

r/R	$\langle L_z \rangle/\hbar$	Physical regime
0.001	0.9997	Nearly circular (spin-1)
0.01	0.997	Thin torus
$\alpha \approx 0.0073$	0.500	**Spin-$\frac{1}{2}$**
0.1	0.91	Moderate torus
0.3	0.73	Thick torus
0.5	0.57	
0.9	0.15	Fat torus (spin $\to$ 0)

Table 1: Angular momentum as a function of aspect ratio.

The transition from integer spin to half-integer spin occurs at $r/R = \alpha$ — the same value fixed by the self-energy self-consistency condition. The angular momentum constraint and the electromagnetic self-energy constraint point to the same geometry.

3.4 Bisection for simultaneous solution

The code performs a two-step procedure:

1. **Find r/R where** $\langle L_z \rangle = \hbar/2$ at fixed R, using bisection on r/R.
2. **Find R where** $E_{\mathbf{total}} = m_e c^2$ at the fixed r/R found in step 1, using bisection on R.

Both constraints are satisfied simultaneously.

4 S4. Topology Catalog

Complete survey of candidate surfaces for the null worldtube:

Surface	Genus	Orient.	$\mathbb{R}^3$	R1	R2	R3	R4	R5	Status
Sphere S^2	0	yes	yes	yes	**no**	**no**	yes	yes	Fails R2, R3
Torus T^2	1	yes	yes	yes	yes	yes	yes	yes	**Model**
Klein bottle	0^*	no	imm.	yes	yes	yes	no*	yes	Forces $p = 2$
Möbius strip	—	no	yes	yes	yes	no	yes	yes	$= T^2$ boundary
Double torus	2	yes	yes	yes	yes	yes	yes	**no***	Curv. unstable
Genus-g	g	yes	yes	yes	yes	yes	yes	**no***	Curv. unstable
Knotted torus	1	yes	yes	yes	yes	yes	yes	yes	Pert. corrections

Table 2: Topology catalog for null worldtube candidates. *Curvature energy exceeds rest mass by factor $\sim 1/(3\pi\alpha) \approx 15$ for genus 2.

The Klein bottle is not a separate surface but provides the *electromagnetic* boundary condition explaining why p must be even. The Möbius strip is the Seifert surface bounded by the $(2, 1)$ torus knot, explaining fermionic statistics. Knotted tori (trefoil, figure-8, etc.) produce mass corrections of order $\alpha^2 m_e \sim$ keV — fine structure, not generation-scale splitting.

5 S5. Klein Bottle EM Boundary Conditions

5.1 The identification

A torus has two independent identifications:

$$\text{Torus:} \quad (x, 0) \sim (x, 2\pi r) \quad \text{and} \quad (0, y) \sim (2\pi R, y). \tag{19}$$

The Klein bottle reverses the second:

$$\text{Klein bottle:} \quad (x, 0) \sim (x, 2\pi r) \quad \text{and} \quad (0, y) \sim (2\pi R, 2\pi r - y). \tag{20}$$

5.2 Effect on EM fields

After one toroidal circuit ($\theta \to \theta + 2\pi$), the poloidal coordinate reverses: $\phi \to 2\pi - \phi$. Under this identification, the electric field component transverse to the circulation direction transforms as $E_\perp \to -E_\perp$ (orientation reversal).

For a standing EM mode $\Psi(\theta, \phi) \propto e^{ip\theta} e^{iq\phi}$, single-valuedness requires:

$$\Psi(\theta + 2\pi, 2\pi - \phi) = \Psi(\theta, \phi), \tag{21}$$

$$e^{i2\pi p} e^{-iq\phi} = e^{iq\phi} \quad \Rightarrow \quad p \in 2\mathbb{Z}. \tag{22}$$

Only **even** toroidal winding numbers produce single-valued electromagnetic modes on the Klein bottle. The minimum is $p = 2$, giving the $(2, 1)$ torus knot as the lightest fermion.

5.3 Mode analysis

Mode (p, q)	Klein bottle	Torus	Statistics
$(1, 1)$	forbidden	allowed	boson
$(2, 1)$	**allowed**	allowed	fermion
$(1, 0)$	forbidden	allowed	trivial
$(3, 1)$	forbidden	allowed	fermion
$(4, 1)$	allowed	allowed	fermion (heavier)

Table 3: Mode compatibility with Klein bottle boundary conditions.

The Klein bottle eliminates all odd-p modes. The remaining modes, with even p, automatically bound Möbius strips and are fermionic. The boson–fermion distinction is thus a consequence of electromagnetic boundary conditions.

6 S6. Gauss–Bonnet Stability Argument

6.1 The theorem

For a closed orientable surface of genus g:

$$\iint K \, dA = 2\pi\chi = 2\pi(2 - 2g), \tag{23}$$

where K is the Gaussian curvature and χ the Euler characteristic.

6.2 Curvature energy

An electromagnetic mode on a curved surface experiences a conformal coupling correction:

$$E^2 \to E^2 + (\hbar c)^2 \frac{K}{6}. \tag{24}$$

For a surface with total area $A = 4\pi^2 Rr$ and average curvature $\langle |K| \rangle = |2-2g|/(2\pi Rr)$:

$$\frac{\delta E_{\text{curv}}}{E} \sim \frac{(\hbar c)^2 |K|}{6E^2} = \frac{|2-2g|}{6\pi \cdot (r/R)} = \frac{|2-2g|}{6\pi\alpha}. \tag{25}$$

6.3 Results by genus

Genus	χ	Flat?	$\delta E/E$	Stable?
0 (sphere)	2	no	—	Fails R2
1 (torus)	0	**yes**	0	**Stable**
2	-2	no	$1/(3\pi\alpha) \approx 14.6$	Unstable
3	-4	no	$2/(3\pi\alpha) \approx 29.1$	Unstable
g	$2-2g$	no	$(g-1)/(3\pi\alpha)$	Unstable

Table 4: Curvature stability by genus.

The curvature instability ratio is **independent of particle mass**: it depends only on the aspect ratio $r/R = \alpha$ and the genus. Any particle — electron, muon, tau, quark — on a genus-2 surface with $r/R = \alpha$ has curvature energy exceeding rest mass by a factor of $\sim$15. The torus is uniquely stable.

For a genus-2 surface to be marginally stable ($\delta E/E < 1$), it would need $r/R > 1/(3\pi) \approx 0.106$ — fifteen times the electron's aspect ratio $\alpha \approx 0.0073$.

7 S7. Retarded Field Structure

The self-interaction of the circulating photon involves retarded fields — the electromagnetic field at each point on the curve due to all other points, evaluated at the retarded time.

For each point i on the curve, the retarded point j satisfies:

$$|\mathbf{r}_i - \mathbf{r}_j| = c(t_i - t_j), \tag{26}$$

with periodic wrapping on the closed worldtube (t is defined modulo T).

At the electron's self-consistent geometry:

- Mean retarded distance: $\langle |\mathbf{r}_i - \mathbf{r}_j| \rangle \sim R$ (comparable to major radius)
- Mean retarded delay: $\langle t_i - t_j \rangle / T \sim 0.4$ (roughly 40% of the circulation period)
- Retarded distance fluctuation: $\sigma_d / \langle d \rangle \sim 0.3$ (moderate variation around the loop)

The retarded field structure confirms that the photon's self-interaction is dominated by the toroidal-scale geometry (R), not the tube-scale geometry (r). This justifies the use of the Neumann inductance formula, which captures the dominant $O(R)$ contribution.

7.1 Physical interpretation

The retarded field calculation reveals that the photon "sees" itself approximately 40% of a circulation period in the past. For the electron, this delay is $\Delta t \approx 0.4T \approx 3.3 \times 10^{-21}$ s. The retarded self-field provides the restoring mechanism: the photon's electromagnetic field, arriving from its own past trajectory, confines the photon to the torus. This is self-consistent because the field and the source are the same object — the photon creates the field that confines it.

The mean retarded distance $\langle d \rangle \sim R$ means the dominant self-interaction is toroidal (between opposite sides of the torus hole), not poloidal (between nearby segments of the tube). This hierarchy — toroidal interaction at scale R dominant over poloidal interaction at scale r — is what allows the self-energy to be treated as a perturbation on the circulation energy.

8 S8. Effective Potential $V_{\text{eff}}(R)$

The total energy as a function of torus major radius, with $r/R = \alpha$ held fixed:

$$V_{\text{eff}}(R) = E_{\text{circ}}(R) + U_{\text{EM}}(R) = \frac{2\pi\hbar c}{L(R)} + \frac{\alpha[\ln(8/\alpha) - 2]}{p\pi} \cdot \frac{2\pi\hbar c}{L(R)}. \tag{27}$$

Since both terms scale as $1/L(R) \propto 1/R$, the potential is **monotonically decreasing**: $V_{\text{eff}} \propto 1/R$. There are no local minima.

This means a single isolated torus has no preferred radius — the system wants to expand indefinitely (lowering its energy by increasing R). Three consequences follow:

1. **Single-particle stability is topological**, not energetic. The torus topology cannot be continuously deformed away; the photon is topologically trapped. The radius R is not fixed by a potential minimum but by external constraints (the Koide quantization condition for three linked tori).

2. **The Bohr atom analogy.** The Coulomb potential $V(r) = -ke^2/r$ is also monotonic (no minimum in the radial potential without the centrifugal barrier). Bohr's quantization condition $2\pi r = n\lambda_{\text{dB}}$ selects discrete radii from the continuum. Similarly, the Koide angle $\theta_K = (6\pi + 2)/9$ selects three discrete torus radii.

3. **Quantum corrections** — running coupling $\alpha(\mu)$, Casimir energy, and Euler–Heisenberg nonlinear QED — are perturbative corrections of order α or smaller. They do not create potential wells. The monotonic $1/R$ shape persists through all corrections computed to date.

Self-consistent radii:

Particle	Mass (MeV)	R (fm)	r (fm)	R/λ_C
e	0.511	194.2	1.42	0.503
μ	105.66	0.939	0.00685	0.00243
τ	1776.9	0.0558	0.000407	0.000144

Table 5: Self-consistent radii for the three charged leptons.

8.1 Quantum corrections to V_{eff}

Three categories of quantum corrections have been evaluated:

Running coupling. One-loop QED gives $\alpha(\mu) = \alpha/[1 - (2\alpha/3\pi)\ln(\mu/m_e)]$. At the electron's torus radius, $\mu = \hbar c/R_e \approx 1$ MeV, this gives $\Delta\alpha/\alpha \approx 0.0004\%$ — negligible. Running coupling becomes significant only at the muon and tau radii, where it contributes corrections of order α^2.

Casimir energy. For a massless field on a torus with periods $2\pi R$ and $2\pi r$, the Casimir energy scales as $E_{\text{Casimir}} \sim (\hbar c/R) f(r/R)$. For the thin torus ($r/R = \alpha$), this is $O(\alpha)$ relative to E_{circ} — the same order as the self-energy already computed and included.

Euler–Heisenberg (nonlinear QED). In strong electromagnetic fields, photon–photon scattering generates an effective four-photon interaction. The correction scales as $\delta E_{\text{EH}}/E \sim \alpha^2 (r_{\text{class}}/R)^4 \approx 5 \times 10^{-5}$ for the electron — entirely negligible.

All quantum corrections are at most $O(\alpha)$, confirming that the monotonic $1/R$ shape of V_{eff} is robust. The stability of individual generations must therefore come from the Koide quantization condition, not from potential wells in $V_{\text{eff}}(R)$.

9 S9. Comparison Tables

9.1 NWT vs. Bohr model

Aspect	Bohr model	Null Worldtube
Orbiting entity	Electron (massive)	Photon (massless)
Central force	Coulomb $-ke^2/r$	Self-interaction U_{EM}
Quantization	$2\pi r = n\lambda$	$\theta_K = (6\pi + 2)/9$
Potential shape	Monotonic $1/r$	Monotonic $1/R$
Stability mechanism	Resonance	Topology + Koide
Ground state	$n = 1$	Electron
Excited states	$n = 2, 3, \ldots$	μ, τ (Koide rotation)
Coupling constant	Input (α)	Output (r/R)
Spin	Added *post hoc*	Emergent (§IV)

Table 6: Comparison of the Bohr model and the Null Worldtube model.

9.2 NWT vs. Schrödinger quantum mechanics

Aspect	Schrödinger QM	Null Worldtube
Electron described by	Wavefunction $\psi(\mathbf{r}, t)$	Null curve on torus
Mass	Input parameter	Emergent: $m = h/(cL)$
Spin	Added via Pauli matrices	Emergent: $L_z = \hbar/2$
Charge	Input parameter	Topological invariant
α	Input parameter	Geometric ratio r/R
Statistics	Spin-statistics axiom	Möbius surface topology
Self-energy	Divergent (needs renorm.)	Finite (UV cutoff at r)
Generations	Not explained	Koide angle θ_K
SM params	26 inputs	3 inputs + 3 integers

Table 7: Comparison of Schrödinger QM and the Null Worldtube model.

9.3 NWT vs. Dirac equation

Aspect	Dirac equation	Null Worldtube
Spinor components	4 (postulated)	2 circ. $\times$ 2 proj.
Zitterbewegung	$\omega = 2mc^2/\hbar$ (formal)	$\omega = 2\pi/T_{\text{tor}}$ (literal)
Antiparticles	Negative-energy sea	Reversed circulation
g-factor	$g = 2$ (exact, tree-level)	$g \approx 2$ (from circ. geom.)
Anomalous moment	$a_e = \alpha/(2\pi)$ (Schwinger)	Self-energy corr. to L_z
CPT symmetry	Theorem from QFT axioms	Geometric: C = rev., P = mirr., T = t-rev.
Spin–orbit coupling	Relativistic correction	Cross-term tor./pol. motion

Table 8: Comparison of the Dirac equation and the Null Worldtube model.

9.4 Key numerical values

Quantity	Symbol	Value	Source
Fine-structure constant	α	7.297×10^{-3}	Input / geometric ratio
Electron mass	m_e	0.5110 MeV/c^2	Self-consistency target
Major radius	R_e	194.2 fm	Bisection
Minor radius	r_e	1.42 fm	αR_e
Compton wavelength	λ_C	386.2 fm	$\hbar/(m_e c)$
R_e/λ_C	—	0.503	$\approx 1/2$
Path length	L	2442 fm	$\approx 4\pi R_e$
Circulation period	T	8.1×10^{-21} s	L/c
Toroidal period	T_{tor}	4.1×10^{-21} s	$T/p = T/2$
Zitterbewegung period	T_{zb}	4.1×10^{-21} s	$2\pi\hbar/(2m_e c^2)$
Current	I	19.7 A	ec/L
Neumann inductance	$\mathcal{L}$	1.2×10^{-18} H	$\mu_0 R[\ln(8R/r) - 2]$
Logarithmic factor	$\ln(8/\alpha) - 2$	4.99	Geometric constant
Self-energy fraction	$U_{\text{EM}}/E_{\text{total}}$	0.5%	$\alpha \ln(8/\alpha)/(2\pi p)$
Torus mode count	—	13	Weinberg denominator

Table 9: Key numerical values for the torus electron.

References

[1] J. D. Jackson, *Classical Electrodynamics*, 3rd ed. (Wiley, 1999).